

Mobile Application Development

Practical No. 21

Develop a program to implement a broadcast receiver to switch between different modes like Airplane mode/Silent Mode/Loud Mode.

Silent Mode:-

1. activity_main.xml :-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/textStatus"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Silent Mode Status"
        android:textSize="20sp"
        android:padding="10dp"/>

    <Switch
        android:id="@+id/switchSilent"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Toggle Silent Mode"/>

</LinearLayout>
```

2. MainActivity.java

```
package com.example.p21;

import android.content.*;
import android.media.AudioManager;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    AudioManager audioManager;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```

Switch s = findViewById(R.id.switchSilent);
TextView t = findViewById(R.id.textStatus);

audioManager = (AudioManager) getSystemService(Context.AUDIO_SERVICE);

// □ Switch for Silent ON/OFF
s.setOnCheckedChangeListener((btn, check) -> {
    if (check) {
        audioManager.setRingerMode(AudioManager.RINGER_MODE_SILENT);
        t.setText("Silent ON");
    } else {
        audioManager.setRingerMode(AudioManager.RINGER_MODE_NORMAL);
        t.setText("Normal ON");
    }
});

// □ Simple Broadcast Receiver
registerReceiver(new BroadcastReceiver() {
    @Override
    public void onReceive(Context c, Intent i) {
        Toast.makeText(c, "Mode Changed", Toast.LENGTH_SHORT).show();
    }
}, new IntentFilter(AudioManager.RINGER_MODE_CHANGED_ACTION));
}
}
3.AndroidManifest.xml

```

```

<uses-permission android:name="android.permission.MODIFY_AUDIO_SETTINGS"/>

```

- **Output:-**

